

Lecture 9: The Quotient Rule; Derivatives of the Other Trigonometric Functions

GOAL: Be able to differentiate functions of the form $\frac{f(x)}{g(x)}$. Use this to derive the derivatives of all 6 trigonometric functions.

Summary:

$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$		
$(\sin x)' = \cos x$	$(\cos x)' = -\sin x$	$(\tan x)' = \sec^2 x$
$(\sec x)' = \sec x \tan x$	$(\csc x)' = -\csc x \cot x$	$(\cot x)' = -\csc^2 x$

NOTE: $\left[\frac{1}{x}\right]' = \frac{-1}{x^2} \neq 0 = \frac{[1]'}{[x]^2}$

$$\left[\frac{f(x)}{g(x)}\right]' = \left[f(x) \cdot \frac{1}{g(x)}\right]' = f'(x) \cdot \frac{1}{g(x)} + f(x) \left[\frac{1}{g(x)}\right]'$$

Q: What's $\left[\frac{1}{g(x)}\right]'$?

A: $\left[\frac{1}{g(x)}\right]' = \frac{-g'(x)}{[g(x)]^2}$

$$= f'(x) \cdot \frac{1}{g(x)} + f(x) \left[\frac{-g'(x)}{[g(x)]^2}\right]$$

$$= \frac{f'(x)g(x)}{g(x)g(x)} - \frac{f(x)g'(x)}{[g(x)]^2}$$

Theorem (Quotient Rule) Let f and g be diff fens

w/ $g(x) \neq 0$. Then,

$$\left[\frac{\overset{\text{"high"}}{f(x)}}{\underset{\text{"low"}}{g(x)}}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

"low d-high, high d-low, square the low, away we go."

Example 1 $\left[\frac{1}{x}\right]' = \frac{-1}{x^2}$

$f \rightarrow \textcircled{1}$
 $g \rightarrow \textcircled{x}$

$$\left[\frac{\textcircled{1}}{\textcircled{x}}\right]' = \frac{[1]' \cdot x - 1 \cdot [x]'}{x^2} = \frac{0 - 1}{x^2} = -\frac{1}{x^2}$$

NOTE: This shows the power rule is true for x^n where n is any integer.

$x^n = \frac{1}{x^{\textcircled{-n}}}$ $\leftarrow$ positive

Example 2 Find y' if $y = \frac{\textcircled{\sin x}}{\textcircled{x^2 + x}}$ $\leftarrow f$ $\leftarrow g$

$f(x) = \sin x \rightarrow f'(x) = \cos x$

$g(x) = x^2 + x \rightarrow g'(x) = 2x + 1$

$$y' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2} = \frac{[\cos x][x^2 + x] - [\sin x][2x + 1]}{[x^2 + x]^2}$$

Example 3 Find y' if $y = \frac{2e^t}{\textcircled{e^t - 5}}$ $\leftarrow f$ $\leftarrow g$

$$y' = \frac{[2e^t]'(e^t - 5) - (2e^t)[e^t - 5]'}{(e^t - 5)^2}$$

$$= \frac{2e^t(e^t - 5) - 2e^t(e^t)}{(e^t - 5)^2} = \frac{2e^{2t} - 10e^t - 2e^{2t}}{(e^t - 5)^2}$$

$$= \frac{-10e^t}{(e^t - 5)^2} = \frac{-10e^t}{e^{2t} - 10e^t + 25}$$

Example 4 Let a be a constant (parameter)

$$\begin{aligned} f \left[\frac{g}{x-a} \right]' &= \frac{[x^2 - a^2]'(x-a) - (x^2 - a^2)[x-a]'}{(x-a)^2} \\ &= \frac{2x(x-a) - (x^2 - a^2)}{(x-a)^2} \quad \leftarrow \text{You can stop here} \\ &= \frac{2x^2 - 2ax - x^2 + a^2}{(x-a)^2} \end{aligned}$$

$$= \frac{x^2 - 2ax + a^2}{(x-a)^2} = \frac{(x-a)^2}{(x-a)^2} = 1$$

Derivatives of Trig Functions

Recall $[\sin x]' = \cos x$ and $[\cos x]' = -\sin x$

Tangent

$$\begin{aligned} [\tan x]' &= \left[\frac{\sin x}{\cos x} \right]' = \frac{[\sin x]' \cos x - \sin x [\cos x]'}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \boxed{\sec^2 x} \end{aligned}$$

Example 5 If $y = \cos x \tan x$, what is y' ?

$$y' = [\cos x]' \tan x + \cos x [\tan x]'$$

You can stop here

$$= -\sin x \tan x + \cos x [\sec^2 x]$$

$$= -\sin x \cdot \frac{\sin x}{\cos x} + \cos x \cdot \frac{1}{\cos^2 x}$$

$$= \frac{-\sin^2 x}{\cos x} + \frac{1}{\cos x} = \frac{1 - \sin^2 x}{\cos x}$$

$$\sin^2 x + \cos^2 x = 1 \Rightarrow \cos^2 x = 1 - \sin^2 x$$

$$= \frac{\cos^2 x}{\cos x} = \cos x$$

Cotangent

$$[\cot x]' = \left[\frac{\cos x}{\sin x} \right]' = \frac{(-\sin x) \sin x - \cos x (\cos x)}{\sin^2 x}$$
$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = \frac{-1}{\sin^2 x} = \boxed{-\csc^2 x}$$

Example 6 Find y' if $y = e^x \cot x$

$$y' = [e^x]' \cot x + e^x [\cot x]'$$
$$= e^x \cot x + e^x (-\csc^2 x)$$

$$= e^x [\cot x - \csc^2 x]$$

Secant

$$[\sec x]' = \left[\frac{1}{\cos x} \right]' = \frac{[1]' \cos x - 1 \cdot [\cos x]'}{\cos^2 x}$$

$$= \frac{0 + \sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$$

$$= \boxed{\sec x \tan x}$$

Example 7 What's y' if $y = \sec^2 x = \underbrace{[\sec x]}_f \underbrace{[\sec x]}_g$

$$y' = [\sec x]' \sec x + \sec x [\sec x]'$$

$$= (\sec x \tan x) \sec x + \sec x (\sec x \tan x)$$

$$= \underbrace{2 \sec x}_{\frac{d}{d(\sec x)}[(\sec x)^2]} \underbrace{(\sec x \tan x)}_{\frac{d}{dx}(\sec x)}$$

Cosecant

$$[\csc x]' = \left[\frac{1}{\sin x} \right]' = \frac{0 \cdot \sin x - 1 \cdot \cos x}{\sin^2 x}$$

$$= \frac{-\cos x}{\sin^2 x} = -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = \boxed{-\csc x \cot x}$$

Example 9 Find $h'(7)$ if $h(x) = \frac{x^2 - 14}{10 - x}$

$f(x) = x^2 - 14$ $g(x) = 10 - x$

$$f(7) = 49 - 14 = 35$$

$$f'(x) = 2x$$

$$f'(7) = 14$$

$$\left\{ \begin{array}{l} g(7) = 10 - 7 = 3 \\ g'(x) = -1 \\ g'(7) = -1 \\ [g(7)]^2 = 9 \end{array} \right.$$

$$h'(7) = \frac{f'(7)g(7) - f(7)g'(7)}{[g(7)]^2} = \frac{14 \cdot 3 - 35 \cdot (-1)}{9}$$

$$= \frac{42 + 35}{9} = \frac{77}{9}$$

Example 8 Find y' if $y = x^5 \csc x$

$$\begin{aligned} y' &= 5x^4 \csc x + x^5 [\csc x]' \\ &= 5x^4 \csc x + x^5 [-\csc x \cot x] \\ &= x^4 \csc x [5 - x \cot x] \end{aligned}$$